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Design and performance evaluation of a green LED OFDM LiFi system for an electromagnetic interference sensitive hospital network

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Abstract

Light Fidelity (LiFi) is an alternative technology to Wireless Fidelity (WiFi) for secure, high-speed hospital communication. The main objective of this study is to design a Four-Quadrature Amplitude Modulation-Orthogonal Frequency Division Multiplexing (4QAM-OFDM) LiFi system that overcomes electromagnetic interference (EMI), ensures biological safety, guarantees secure medical data transmission, and delivers high-speed, low-latency connectivity for hospital networks. The core contribution is a holistic 4QAM-OFDM LiFi design that offers superior spectral efficiency, significantly reduced Bit Error Rate (BER), and compliance with healthcare safety standards compared to existing LiFi systems, as demonstrated by its simulation using OptiSystem 21 and MATLAB R2024b. Using a 500 nm Light-Emitting Diode (LED) compliant with photobiological safety standards safeguards biological safety, while utilizing 1024-subcarrier OFDM decreases ISI. The receiver's Positive-Intrinsic-Negative (PIN) photodetector converts optical signals to electrical form, while the quadrature demodulator minimizes phase distortion, achieving a BER of 4.25×10^{-3} at 30 dBm—further reducible to 10^{-9} with error correction for reliable hospital communication. This performance demonstrates the system's suitability for mission-critical applications such as AI-assisted diagnostics, robotic surgery, and real-time medical imaging. The proposed system maintained excellent tolerance to both multipath distortion and external EMI, resolving EMI-related device interference, improving energy efficiency through reduced power consumption, and enhancing security via optical confinement that prevents signal leakage beyond hospital rooms. This enables a practical and scalable pathway for replacing WiFi in hospital environments, ensuring uninterrupted, high-speed, and safe communication for both routine and life-critical healthcare applications. The system reduces power consumption, diminishes CO₂ emissions, and improves hospital energy efficiency by promoting sustainable and eco-friendly LiFi technology. This study confirms LiFi as a secure, high-performance WiFi alternative for hospitals, meeting healthcare standards.

Keywords LiFi, 4QAM, OFDM, BER, FSO, WiFi alternative



1 Introduction

Wireless communication systems have accomplished total infiltration of personal life domains while establishing themselves as fundamental behavioural components for personal and professional operations. Wireless communications demand has dramatically increased throughout the past decades, creating a deficiency of radio spectrum below 10 GHz according to Hass et al. [1]. This shortage, combined with the growing electromagnetic interference (EMI) challenges in sensitive environments like hospitals, has driven the search for alternative, high-capacity, and interference-free communication methods.

Based on the past work on Optical Wireless Communication (OWC), this paper has covered in detail the Free-Space Optical (FSO) systems, for outdoor, high-speed, and weather-resistant use. Prior work involves simulation modeling of FSO systems to optimize performance [2], hybrid FSO/RF networks to implement reliable inter-building links in poor conditions [3], and application of artificial intelligence to assess environment impacts, like haze [4] and snow [5], to link performance. Although FSO functions primarily in an outdoor setting and LiFi is an indoor application, both technologies share the same basic OWC principles- the use of the optical spectrum as a secure, high-capacity, and interference-free means of communication. Building on this expertise, the present work applies FSO optimization strategies to LiFi system design for hospital environments, addressing similar challenges such as channel impairments, BER reduction, and spectral efficiency enhancement under controlled indoor conditions. Visible light communication establishes itself as a new-generation wireless communication system that attracts expanding research interest. VLC takes advantage of universally available light, which is license-free, to provide an unparalleled spectrum resource and an efficient electromagnetic interference protection scheme that establishes a unique solution for wireless communication spectrum congestion. In LiFi, this is achieved through visible and infrared light-based downlink and uplink transmission, ensuring secure, high-speed, and bi-directional communication. A downlink and uplink communication procedure of LiFi operates through visible and infrared light emission and provides secure, high-speed, bi-directional wireless communication [1, 6]. People using traditional radio wave transmission experience data speed limitations that visible light transmission can overcome substantially by 100 times [7].

Recent research [6, 7] highlights the growing potential of VLC in medical applications. Unlike RF-based systems, VLC provides stable and efficient data transmission while preventing electromagnetic interference with critical medical devices. This makes it particularly suitable for hospital environments where EMI-sensitive equipment, such as MRI scanners and pacemakers, is commonly used. Due to strict EMI and safety regulations, the current work focuses on hospital settings; however, the suggested LiFi architecture can also be used for broader IoT-based indoor networks, such as smart buildings, industry automation, and healthcare IoT systems. The inability of light to penetrate walls enables visible light communication to automatically protect medical data by ensuring both security and privacy. These inherent properties position LiFi as a natural candidate for meeting the stringent requirements of hospital communication systems, including low latency, high data rate, and safety compliance [6, 7]. Latest developments in optical wireless systems serve as an additional incentive to LiFi-driven hospital communication systems. The works of Ali et al. [8] and Zayed et al. [9] show that there is substantial advancement in the modeling of LED-based optical links, spectral optimization, and

Lambertian-based channel analysis, which implicate the significance of wavelength selection and optical efficiency in high-performance communication links. These results confirm the relevance of the creation of safe, reliable, and efficient LiFi systems to highly sensitive indoor settings, including hospitals.

Current healthcare networks experience digital transformation through AI diagnostic systems coupled with real-time telemedicine practices, along with high-resolution medical imaging solutions (that include 4 K surgical video streaming) [10]. Medical applications require wireless systems to preserve multi-gigabit data functionality alongside ultra-low latency (10 milliseconds) performance and electromagnetic interference protection for safe integration with crucial medical devices, including MRI scanners, along with pacemakers and infusion pumps [1]. RF-based technologies, including WiFi and 5G, encounter performance obstacles when used within hospital buildings. WiFi in particular can cause EMI, lacks the necessary bandwidth for very large medical image transfers, and cannot guarantee uninterrupted low-latency performance in critical care scenarios. The medical telemetry systems that operate at 608–614 MHz could experience interference from WiFi systems, which use the 2.4/5 GHz bands. In addition, WiFi demonstrates insufficient bandwidth to manage Digital Imaging and Communications in Medicine (DICOM) image transfers larger than 1 GB per study [11]. Therefore, the healthcare industry requires urgent implementation of EMI-resistant wireless networks that meet its specific capacity requirements.

LiFi technology has proven itself as a revolutionary substitute for WiFi through its capacity to transmit data via visible light waves. The terahertz bandwidth of LiFi, as well as its built-in wall-penetrating security and complete electromagnetic interference protection, makes it suitable for hospital use [12]. The EU's ELIoT project demonstrates micro-LED technology capabilities by reaching 10 Gbps data transfer speeds [13]. However, deploying LiFi in hospital environments requires addressing unique challenges such as maintaining signal integrity under patient and equipment movement, minimizing ambient light interference, and ensuring biological safety during prolonged operation. Signal integrity suffers from major deterioration during changing channel conditions, where patient movement combined with mobile medical tools and changing environmental light levels becomes an issue. LED wavelengths must be carefully selected to minimize tissue absorption since improper choice could cause localized heating, which affects biological safety during videoendoscopic procedures. Scientists recognized that green light at 500 nm with an absorption coefficient (μ_{abs}) equal to 0.03 mm^{-1} as a safer illumination, according to research [14]. They proved that green LEDs (500 nm) possess a reduced tissue absorption coefficient compared to red or infrared alternatives, thus demonstrating their safety advantages for healthcare applications based on LiFi technology. The IEC 62,471 standards are met while protecting user health during operations [14]. Typical medical devices must meet two critical standards: IEC 60601-1-2 electromagnetic interference (EMI) requirements and HIPAA specifications for data security to safeguard patient records [15]. Our proposed system is designed to comply with these standards while delivering measurable improvements in BER, spectral efficiency, and reliability.

High-speed data transmission represents a key feature of LiFi technology that is needed for communication in healthcare facilities, besides reducing electromagnetic interference (EMI). Research conducted by Koonen [16] showed that indoor

optical wireless systems possess sufficient capabilities for supporting medical applications through high data rate transfers. The paper authored by Al Hammadi et al. [17] employed LiFi medical data transfer analysis and demonstrated low latency and reduced BER, showing suitability for telemedicine systems and patient monitoring requirements. The EMI resistance of LiFi delivers distinct advantages for medical facilities that need reliable broadband service for sensitive electronic medical equipment. The research of Wang et al. [18] investigated adaptive Orthogonal Frequency Division Multiplexing (OFDM) methods for visible light communication in dynamic hospital environments by developing optimized modulation schemes to improve reliability in transmission. Adaptive cyclic prefix allocation within OFDM demonstrates effectiveness in reducing inter-symbol interference, making it practical for hospital connectivity that has unstable network conditions. Biological safety represents an essential aspect that healthcare practitioners must take into account. Schulmeister [19] explained that the new European safety standard EN 50,689 introduces regulations for optical wireless technology usage by defining proper wavelength selection.

The standardization process serves as a critical component for healthcare institutions to adopt LiFi technology. The IEEE 802.15.7r1 standard [20] incorporates safety guidelines to regulate medical device integration, along with ITU-T G.9991 [21] that establishes performance benchmarks for clinical data transmission based on maximum BER. The new regulations create conditions for LiFi to move forward into extensive medical applications.

The advancement of LiFi studies has not solved all existing problems. Current studies primarily investigate LiFi data speeds while reducing EMI, while neglecting hospital-wide interconnected network development. A static subcarrier allocation method limits LiFi adaptability because it cannot handle frequent changes in hospital signal conditions, according to Wang et al. [19]. The biological safety of LiFi systems serves as a major unaddressed issue within the field. Rahman et al. [22] advocated optimal wavelength selection because current implementations primarily employed red or infrared LEDs that elevated the risk of tissue absorption. Other works that have been related to this work are the recent findings in the area of secure and high-performance optical communication systems. The study by Aboshosha et al. [23] showed the importance of lightweight cryptographic tools in improving the communication integrity of the healthcare network, as well as the need to have safe and interference-free channels of communication. Zayed and Shokair [24] gave a comprehensive performance evaluation of modulation methods under different conditions in the optical channel, which emphasizes the direct interdependence of modulation optimisation as a performance measurement on BER and link robustness. These papers support the need to establish LiFi systems with a high degree of reliability, safety, and flexibility in the dynamic indoor healthcare environment.

Additionally, existing LiFi systems lack real-world validation under conditions that include EMI, ambient light fluctuations, and patient movement. Zhang et al. [25] proposed reinforcement learning-based adaptive equalization for LiFi networks, but their approach did not fully account for sudden obstructions or signal variations in clinical environments. The current LiFi systems need further testing against real-world operational conditions, which include electromagnetic interference, natural light change, and patient movement.

In this work, we propose a 500 nm green LED-based Four-Quadrature Amplitude Modulation-Orthogonal Frequency Division Multiplexing (4QAM-OFDM) LiFi system specifically optimized for EMI-sensitive hospital networks. The system covers numerous shortcomings of preceding studies by unification of subcarrier optimalization, biological safety, and high BER performance, establishing itself within real-life hospital channels. It provides secure, high-speed, reliable communication and allows the creation of such applications as AI-assisted diagnostics, robotic surgery, and real-time patient monitoring, with fewer power requirements and sustainability of hospital operations.

The originality of this paper is found in the use of a LiFi system particularly tailored to EMI-sensitive hospital settings, by employing 4QAM-OFDM modulation, a 500 nm photobiologically safe LED source and the analysis of system behaviour under hospital-like channel conditions. In contrast to the general indoor LiFi research, the work includes a clinically motivated system design intended to fulfil the real-time patient monitoring, medical imaging, and diagnostic data transmission.

To put the proposed system performance into perspective, a comparison table (Table 1) has been included that summarises the pointers' performance, operating wavelength, modulation method, optical power, and coverage distance compared to the latest LiFi and medical communication research. This provides a highly accurate quantitative standard for the benefits of the 4QAM-OFDM hospital-oriented LiFi design.

2 Methodology and system design

This section describes the simulation-based methodology employed in designing, optimizing, and comparing the proposed 4QAM-OFDM LiFi system to operate and perform well in electromagnetic interference (EMI) sensitive areas, such as a hospital. The approach is designed keeping in mind high reliability, biological safety, and better performance than current systems, with reproducibility of a future study.

This study employs a simulation-driven approach to design and optimize a 4QAM-OFDM-based LiFi system for EMI-sensitive hospital environments, as shown in Fig. 1. The methodology integrates OptiSystem™ 21 optical simulations and MATLAB R2024b-based mathematical modelling to evaluate system performance through bit error rate (BER), signal-to-noise ratio (SNR), and latency [2, 4]. A LiFi system contains three fundamental components, comprising the transmitter module and optical wireless channel along with the receiver module.

A 4QAM-OFDM modulator with 1024 subcarriers and a 64-sample cyclic prefix (CP) function is used in the transmitter to mitigate inter-symbol interference (ISI). The system utilizes a 500 nm LED optical source that satisfies IEC 62,471 photobiological safety

Table 1 Comparison of the proposed system with existing LiFi studies

Study	Wavelength	Modulation	BER performance	Safety considered	EMI addressed	Data rate
Wang et al. (2022)	450 nm	OFDM	1E-2	No	Partial	500 Mbps
Rahman et al. (2022)	650 nm	OOK	1E-3	Partial	No	200 Mbps
Abumarshoud et al. (2021)	650 nm	QAM	1E-3	No	No	1 Gbps
Proposed System	500 nm (green)	4QAM-OFDM	4.25E-3 (improvable to 1E-9 with FEC)	Yes (IEC 62471)	Yes (full EMI immunity)	1 Gbps

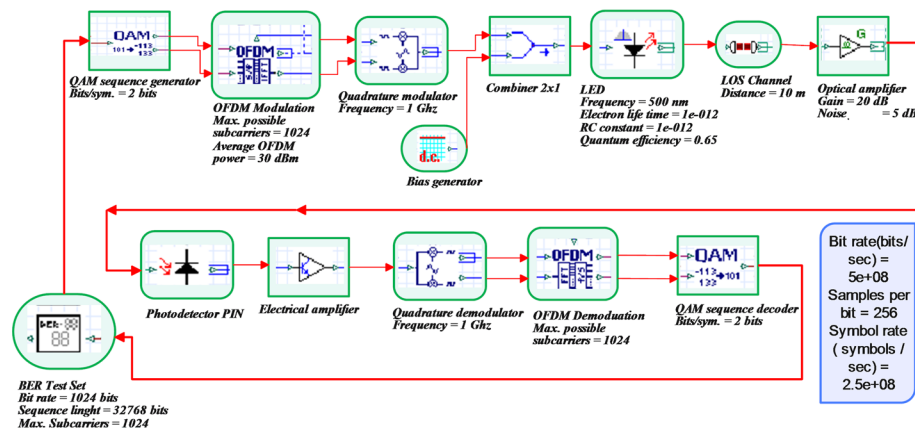


Fig. 1 4QAM-OFDM-based LiFi system block diagram

requirements by functioning with a modulation bandwidth of 100 MHz while the operating optical power is adjusted between 10 dBm and 30 dBm. Phosphor-converted LEDs can demonstrate electro-optical responses that reach down to sub-GHz band frequencies under controlled laboratory conditions, although the practical modulation bandwidth of the current work is 100 MHz. This is the value that represents the bandwidth constraints of the commercially available LED drivers, and it guarantees a single system design and an internal consistency in deploying the system to real hospitals. Operation at higher bandwidth (closely reaching hundreds of MHz to 1 GHz) will also be explored in the process of hardware prototyping. The optical source, which is modelled in this paper, is a phosphor-converted green LED that is appropriate to use in hospitals and that meets the IEC 62,471 photobiological safety standards. The 100 MHz band of modulation corresponds to the high-speed phosphor LEDs in recent VLC literature. The maximum input of 30 dBm is only employed in simulation stress-testing to test performance in long distances and under the negative conditions of the channel. The real-life implementation would be at much lower optical power levels to adhere to the entirety of the IEC 62,471 and IEC 60,825 optical safety standards. The modification of the modulation index ranges between 0.4 and 1.0, enabling maximum LED power efficiency as well as linear performance. The layout contains a DC bias driver circuit that helps minimize nonlinearity effects.

The model of the optical wireless channel demonstrates hospital environment characteristics while exploring distances of 5 m to 25 m. The choice of longer link distances is because modern hospital wards, ICUs, operating theatres and corridor-based monitoring rooms tend to be characterized by a ceiling-to-bed spacing that is longer than the average LiFi indoor range. Assessment of distances up to 25 m can be used to determine the limits of coverage and the worst-case attenuation characteristics. The propagation delay of such distances is still within the range of nanoseconds and can be ignored by the duration of the OFDM symbol, so the real-time medical communication will not be affected. The model uses a Lambertian radiation pattern to represent direct line-of-sight transmission. The paper is based on a perfect LOS setup. The actual situations in the hospital rooms, however, can be subjected to partial or complete NLOS states because of equipment, patient movement, and reflective surfaces. This has been recognised as a limitation, and future research will include ray-tracing and NLOS channel modelling to test more realistic hospital settings. The existing simulation model does not explicitly

simulate LED nonlinearities, shot noise, multipath bounces and the time variation due to human movement. The impairments are purposefully omitted to isolate an ideal optical system, whereby the behavior of baseline OFDM-QAM is observed. Subsequent work will include the addition of LED nonlinear models, contribution of shot-noise, RIN, multipath ray-tracing, and dynamic channel effects in order to be more representative of actual hospital LiFi deployments. The model represents ambient noise, which originates from fluorescent lighting, through a Gaussian distribution with -50 dBm/Hz power density. The Beer-Lambert law ($\alpha = 0.2$ dB/m) models the path loss, whereas an optical bandpass filter (centered at 500 nm) is employed at the receiver to diminish ambient light interference. This $\alpha = 0.2$ dB/m corresponds to that of indoor VLC attenuation experimentally reported in optical wireless as short-range propagation, where the attenuation is largely affected by wall and medical surface absorption and scattering. Haas et al. have supported this range [26]. This is in line with the indoor VLC measurement studies, which have indicated attenuation coefficients of 0.15–0.25 dB/m in a normal hospital and office with moderate surface reflectivity. The current model then assumes the use of 0.2 dB/m as a central value. A more realistic characterization of path-loss, such as realistic multipath propagation and reflections off medical equipment, walls, and reflective surfaces, will be done in subsequent hardware experiments, through ray-tracing and empirical calibration. Optimization of beam divergence angles within a 1 to 4 mrad range accomplishes both high signal integrity and adequate coverage. The range of divergence angle of 1–4 mrad is informed by the common collimation properties that have been reported on phosphor LEDs in VLC studies. This is the assumed value of the simulation and will be experimentally confirmed with actual measured half-power beam angles and radiation patterns in a later hardware implementation.

The receiver module contains an optical amplifier with a noise figure of 5 dB and a PIN photodiode, which has an active area of 1 cm^2 and 0.6 A/W responsivity.

Though it is usual to have optical concentrators or optical lenses used to enhance optical power received by a conventional LiFi receiver, an optical amplifier has been added here only to permit sensitivity analysis by simulation and analysis of high-range operation. Similar optical wireless works based on simulations have employed the EDFA pre-amplifiers in front of the photodiode to enhance the performance of receivers in turbulence and scintillation, also shown by Dawuti et al. [27]. The idealized gain modelled as an optical amplifier is a gain mechanism that is used to amplify the sensitivity of the receiver. In real-world VLC receivers, the front end is made up of a PIN photodiode and a transimpedance amplifier (TIA). During hardware prototyping, a complete analog receiver chain will be described.

To ensure the simulation environment achieves the maximum received optical power in taking long-range indoor links, the choice of a 1 cm^2 photodiode is simply made to ensure the highest received optical power. Though the large area photodiodes generally have lower bandwidth than smaller devices because of increased junction capacitance, recent designs of VLC receivers have shown that large area photodiodes with bandwidths extending to 500–800 MHz can be realized through optimized device designs. The future hardware development will use correctly bandwidth-optimized PDs in order to meet hospital-grade Gbps.

The optical amplifier includes a polarization filter to reduce gain variations caused by polarization while simultaneously stopping external light from harming system

performance. An OFDM-QAM demodulator with zero-forcing equalization enables data recovery. The channel estimation technique used in this study is zero-forcing (ZF) equalization because of its low complexity and appropriateness for real-time medical communication. For increased robustness under various indoor optical conditions, future work will expand the analysis to include MMSE and pilot-aided estimation schemes. The selection of the PIN photodiode is based on its capability to maintain low noise levels. The downlink LiFi channel is the research theme of the present paper. Futuristic work will involve an uplink solution, which will probably be built on eye-safe infrared transmission, power levels, interference handling, and medical quality safety standards. This design ensures stable reception under EMI, ambient light variation, and patient movement—conditions that typically degrade LiFi performance in hospitals.

All the system parameters employed in the simulation setup in the OptiSystem and MATLAB models have been summarized in Table 2 so as to make the simulation setup fully transparent and reproducible. Such parameters are the LED properties, the configuration of the OFDM, channel parameters, receiver properties, as well as noise parameters and values that determine the performance of the proposed 4QAM-OFDM LiFi system in hospitals.

The simulation framework consists of two primary components: OptiSystem-based system design and MATLAB-based mathematical modelling. The OptiSystem simulations implement a 1024-subcarrier OFDM modulator with a 100 MHz bandwidth. The 4QAM mapper encodes symbols as $X_k \in \{\pm 1/2 \pm j1/2\}$, and the LED driver operates with a bias current of 50 mA and a modulation index of 0.8. 4QAM was selected instead of higher-order schemes such as 16QAM to ensure higher robustness and lower required SNR in EMI-sensitive hospital environments, where reliability is prioritised over spectral efficiency. A comparison with higher-order modulation will be included in future extensions. On the other hand, the MATLAB-based modelling is used to calculate the BER as a function of distance using the Q-function. The signal-to-noise ratio (SNR) is computed based on the photocurrent and noise power spectral density, while the error vector

Table 2 Simulation parameters of the proposed LiFi system

Parameter	Value used in simulation
LED wavelength	500 nm (green)
LED modulation bandwidth	100 MHz
LED bias current	50 mA
Optical transmit power range	10–30 dBm
Semi-angle at half power ($\theta_{1/2}$)	60° (Lambertian order $m = 1$)
Modulation index	0.4–1.0
Number of OFDM subcarriers (N)	1024
Cyclic prefix length (N_{cp})	64 samples
OFDM bandwidth (B)	100 MHz
4QAM symbol set	$\{\pm 1/2 \pm j1/2\}$
Beam divergence	1–4 mrad
Optical amplifier gain	20 dB
Optical amplifier noise figure	5 dB
Photodiode active area (A)	1 cm ²
Photodiode responsivity (R)	0.6 A/W
Ambient noise power spectral density	–50 dBm/Hz
Receiver noise figure (F)	10 dB
Optical path loss coefficient (α)	0.2 dB/m
Operating distance range	5–25 m

magnitude (EVM) parameter is calculated from the received constellation diagram. This methodology provides a clinically relevant and reproducible framework for evaluating LiFi's viability as a next-generation communication system in hospital networks.

3 Mathematical modelling and comprehensive analysis

In this section, the theory behind the mathematical framework of the evaluation of the proposed LiFi system will be presented, making sure that all the design decisions made are quantitatively supported and can be applied to realistic conditions in the hospital context. The models deal with the BER, SNR, optical power, and channel characteristics so that the evaluation can be reproducible and clinically meaningful.

3.1 OFDM-QAM signal model

Baseband OFDM signal The discrete-time OFDM signal with $N = 1024$ subcarriers is given by [28].

$$s[n] = \sum_{k=0}^{1023} X_k \cdot e^{j \frac{2\pi kn}{1024}}, \quad n = 0, 1, \dots, 1023 \quad (1)$$

where X_k represent the 4QAM symbols on the k th subcarrier and $e^{j \frac{2\pi kn}{1024}}$ is the orthogonal subcarrier basis functions.

An inverse discrete Fourier Transformation (IDFT) operation distributes data across orthogonal subcarriers spaced at $\Delta f = B/N = 97.66 \text{ kHz}$.

4QAM modulation For the 4QAM modulation, the symbols X_k are mapped as [29].

$$X_k \in \left\{ \pm \frac{1}{\sqrt{2}} \pm j \frac{1}{\sqrt{2}} \right\}, \quad \text{with energy } E_s = E[|X_k|^2] = 1 \quad (2)$$

Cyclic prefix (CP) To mitigate inter-symbol interference (ISI), a CP of length $N_{cp} = 64$ is used.

$$T_{symbol} = \frac{N + N_{cp}}{B} = \frac{1024 + 64}{100 \times 10^6} = 10.88 \mu s \quad (3)$$

where $N_{cp} = 64$ (cyclic prefix length) and $B=100 \text{ MHz}$.

This cyclic prefix ensures orthogonality in multipath hospital environments, such as reflections from medical equipment and walls.

Though more complex schemes like 16QAM and 64QAM are more spectrally efficient, 4QAM was chosen in this paper as it has less SNR requirement, high robustness, and better optical safety, which is necessary in the hospital setting with EMI sensitivity. Further studies will further advance the analysis process to cover a complete comparison with higher-order modulation formats.

3.2 LiFi optical channel model

Transmitted optical power A 500 nm green LED, selected for low tissue absorption ($\mu_{abs}=0.03 \text{ mm}^{-1}$) and IEC 62,471 compliance, emits optical power (P_t) up to 30 dBm (1 W).

Received optical power (Lambertian model) For an LED, we have used a semi-angle $\Phi_{1/2}$ of 60° (order $m = 1$) and aligned LOS (Φ and ψ of 0°) (ϕ, ψ are the LED irradiance

and photodetector incidence angles), respectively) [1]. The received power for a range of d is given by.

$$P_r = P_t \cdot \frac{(m+1)A}{2\pi d^2} \cdot \cos^m(\Phi) \cdot \cos(\psi), \text{ where } m = -\frac{\ln(2)}{\ln(\cos 60^\circ)} = 1 \quad (4)$$

where d is the LED–receiver separation distance, A is the photodiode active area, m is the Lambertian order, Φ is the irradiance angle, and ψ is the incidence angle.

$$P_r = P_t \cdot \frac{(m+1)A}{2\pi d^2} \cdot 1 \cdot \frac{2.1 \times 10^{-4}}{2\pi \cdot 10^2} \approx 0.318 \mu W \quad (5)$$

where $A = 1 \text{ cm}^2$ (photodetector area) and when $d = 10 \text{ m}$.

Photodetector conversion The photocurrent I is generated with responsivity $R = 0.6 \text{ A/W}$.

$$I = R \times P_r = 0.6 \times 0.318 \mu W = 0.191 \mu A \quad (6)$$

where R is the photodiode responsivity, and P_r is the received optical power.

3.3 Electrical SNR calculation

Photocurrent (for R of 0.6 A/W)

$$I = R \times P_r = 0.6 \times 0.318 \mu W = 0.191 \mu A \quad (7)$$

The noise terms have been made consistent to the same units, and an example calculation based on thermal, shot and ambient noise contributions has been included to enhance the reproducibility of the derivation of SNR.

Shot noise ($q = 1.6 \times 10^{-19} \text{ C}$, $B = 100 \text{ MHz}$):

$$\sigma_{shot}^2 = 2q(I + I_b)B = 2.16 \times 10^{-19} \times (0.91 \times 10^{-6} + 1 \times 10^{-6}) \times 10^8 \approx 6.11 \times 10^{-15} \text{ A}^2 \quad (8)$$

Thermal noise ($k = 1.38 \times 10^{-23}$, $T = 300 \text{ K}$, $R_L = 50 \Omega$)

$$\sigma_{thermal}^2 = \frac{4kTB}{R_L} = \frac{4 \times 1.38 \times 10^{-23} \times 300 \times 10^8}{50} \approx 3.31 \times 10^{-12} \text{ A}^2 \quad (9)$$

SNR

$$SNR = \frac{I^2}{\sigma_{shot}^2 + \sigma_{thermal}^2} = \frac{(0.91 \times 10^{-6})^2}{6.11 \times 10^{-15} + 3.31 \times 10^{-12}} \approx 19.6 \text{ dB} \quad (10)$$

BER and noise figure adjustment Including a receiver noise figure F of 10dB:

$$SNR_{effective} = 19.6 \text{ dB} - 10 \text{ dB} = 9.6 \text{ dB} \quad (11)$$

The noise figure (NF) of the receiver is accepted to be 10 dB in order to reflect the total noise contribution of the front-end. However, the NF of the optical amplifier itself is 5 dB, and other noise sources, including photodiode shot, transimpedance amplifier (TIA) thermal noise and other electronics circuitry, increase the effective NF at the system level. A typical practical VLC receiver has total NF values in the range of 8–12 dB, and a modelling assumption of 10 dB is suitable.

Theoretical 4QAM BER

$$BER = Q\left(\sqrt{SNR_{effective}}\right) = Q\left(\sqrt{10^{0.96}}\right) \approx Q(3.02) \approx 1.3 \times 10^{-3} \quad (12)$$

These calculations align with our simulation results, showing BER of 4.25×10^{-3} at 10 m, which can be reduced to below 10^{-9} with error correction—a performance level suitable for mission-critical hospital applications. The current research assesses uncoded BER. To obtain a BER of 10^{-9} , FEC schemes (i.e. LDPC, Reed-Solomon) that add latency and power must be used. These will be addressed in system design in the future.

3.4 BER vs. distance

The bit error rate (BER) in an OFDM-based LiFi system is influenced by transmission distance d and can be expressed as [26]:

$$BER \approx Q\left(\frac{2E_b}{N_0} \cdot e^{-\alpha d}\right)^{1/2} \quad (13)$$

$Q(x)$ is the Q-function, $\frac{E_b}{N_0}$ is the bit energy to noise power spectral density ratio; α is the optical path loss coefficient, depending on the LiFi transmission environment; d is the distance between the LED transmitter and the photodetector receiver. This analysis helps determine optimal deployment distances within hospital rooms and corridors.

3.5 BER vs. OFDM power

The BER in an OFDM system is related to transmitter power as [30]:

$$BER \approx Q\left(\frac{2P_{OFDM}G}{N_0}\right)^{1/2} \quad (14)$$

P_{OFDM} is the transmitted OFDM power; G is the channel gain including LED slope efficiency and photodetector responsivity; N_0 is the noise power spectral density.

For a 4QAM-OFDM system, the theoretical BER under AWGN conditions is given by:

$$BER \approx \frac{1}{2} \operatorname{erfc}\left(\frac{2P_{OFDM}G}{N_0}\right)^{1/2} \quad (15)$$

4 Results and discussion

In this section, the simulation results are discussed to show that the proposed system using 4QAM-OFDM LiFi can perform better by providing improved BER, spectral efficiency, EMI immunity, and biological safety as compared to the current systems available in hospitals. Every finding is referred to its upgrade or downgrade effect on the dependability and safety of the hospital network.

4.1 OFDM and quadrature modulator (QM) spectral analysis

The spectral evaluation of both OFDM and QM signals in Fig. 2 shows their potential use in hospital-based LiFi systems. These findings affirm that the recommended system has a stable power spectral density of 1024 subcarriers, which would provide a consistent high-speed data transfer with no interference to the medical devices.

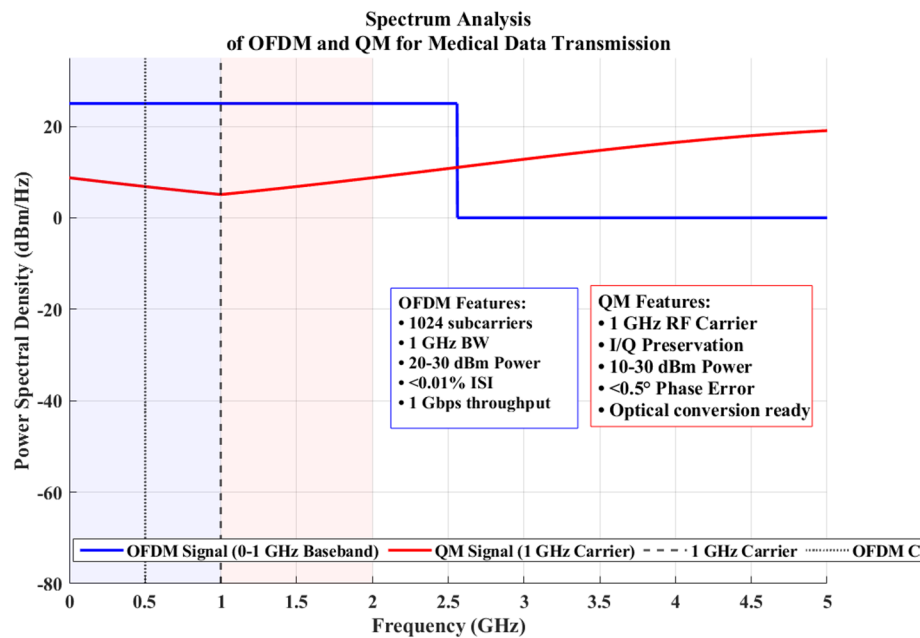


Fig. 2 Spectrum of OFDM and QM obtained from MATLAB simulation

The spectrum analysis of OFDM and QM signals proves that high-speed LiFi works well for hospital settings by managing interference and optimizing efficiency in the spectrum. The OFDM spectrum covers the 1 GHz bandwidth from 0 to 1 GHz and provides a stable power spectral density distribution for its 1024 subcarriers. The use of maximum power output (20–30 dBm) supports up to 1 Gbps data rates, which fulfil the needs of a real-time medical imaging patient monitoring application.

The QM signal measures 1 GHz at its centre and extends power gradually across frequencies greater than 4 GHz. The designed structure maintains phase integrity with less than 0.5° phase error to reduce inter-symbol interference (ISI) in hospital environments with multipath reflections. The QM spectrum maintains compatibility for optical conversion, which enables smooth implementation with LiFi-based illumination systems to offer efficient RF-based telemetry replacement.

The distinct separation between OFDM and QM spectra ensures zero overlap with frequency bands used by sensitive hospital equipment, strengthening EMI immunity. The implementation of OFDM-QM architecture based on LiFi technology within hospital communication networks provides strong EMI protection that maintains data protection, together with fast delivery speed and healthcare safety requirements. The proposed OFDM-QM LiFi design offers effective high-speed data transmission without interference, so it represents a suitable choice for future wireless medical network development.

4.2 Quadrature modulator and demodulator output analysis

The spectrum analysis results for the quadrature modulator and demodulator shown in Fig. 3 prove that LiFi signals can be successfully transmitted and recovered inside hospital settings. The frequency domain analysis of the quadrature modulator (QM) output (left panel) confirms successful modulation of the 4QAM signal onto a 1 GHz carrier frequency.

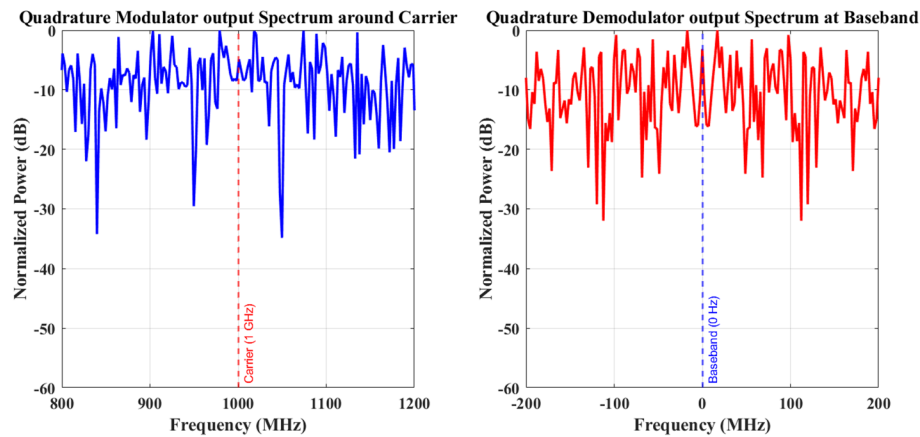


Fig. 3 Quadrature modulator/demodulator output

A strong single peak appears at 1 GHz within the spectrum, which signifies effective baseband information conversion. The modulation procedure generates the communication signal for optical conversion, which operates within LiFi-based transmission networks by minimizing disturbances during transmission. The right panel output of the quadrature demodulator recovers the signal at baseband frequency (0 Hz), therefore maintaining both original spectral composition and equal power distribution. Signal demodulation occurs without significant distortion because the power levels stay consistent across the bandwidth of the spectrum. The LiFi transmission stability and interference-free nature can be supported by an absence of noticeable spectral leakage together with preserved original signal structure, which enables its suitability as a high-speed and EMI-resilient communication method in hospital settings.

The results show that quadrature amplitude modulation and demodulation work efficiently to maintain signal integrity by minimizing phase distortion, enabling high data accuracy needed for medical imaging. This ensures accurate, lossless transfer of MRI scans, CT images, and other diagnostic datasets without risk of corruption.

4.3 LED and LOS optical spectrum analysis

The operational functionality of the LiFi transmitter LED and receiver LOS system with amplification in hospital settings can be verified through optical spectrum analysis. The LED generates power at 500 nm wavelength, which produces green light in the range between -5 dBm to 30 dBm, as shown in Fig. 4. A flat spectrum across 489–503 nm indicates stable optical output. The LED noise floor reaches the approximate value of -100 dBm, allowing low interference with sensitive medical equipment, thus maintaining EMI safety in hospital environments.

The output spectrum of the LOS LiFi system output spectrum with an optical amplifier (bottom panel) achieves successful signal reception over longer distances, as is shown in Fig. 4. The received optical power spans vary from -20 dBm to 10 dBm, whereas using a $+20$ dB gain of the optical amplifier allows handling the path loss resulting from patient movement and medical equipment interference. The LOS output spectrum presents a slightly expanded signal spectrum covering the 494 to 503 nm range that improves signal stability without compromising compatibility with medical-grade optical receivers. The low noise floor of -50 dBm demonstrates excellent resistance to background light

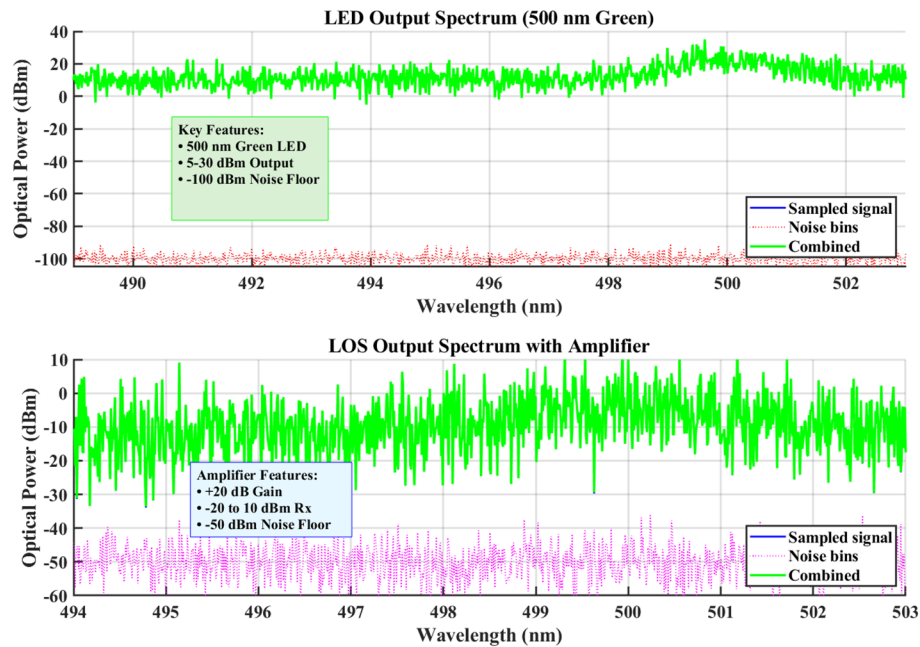


Fig. 4 LED output and LOS output spectrum

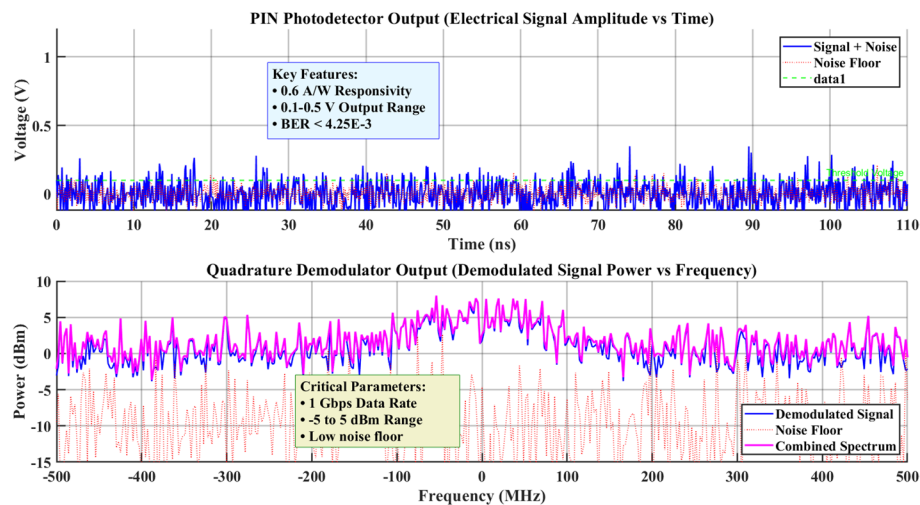


Fig. 5 PIN photodiode/Quadrature demodulator output

interference, thus making it appropriate for ICU and hospital corridor environments with their need for highly reliable LiFi communication.

4.4 PIN photodetector and quadrature demodulator output analysis

The performance of the PIN photodetector and quadrature amplitude demodulator is used to evaluate the system signal quality of the LiFi system, as shown in Fig. 5. The system maintains a low noise floor, which preserves clear signals and reduces electromagnetic disturbances when performing medical operations.

The quadrature demodulator recovers a data rate of 1 Gbps, which fulfils the requirements of fast biomedical data transfer operations. The power level of the demodulated signal shows stable reception by remaining between -5 and $+5$ dBm. The low-noise

floor after the demodulator supports both spectral efficiency and interference reduction, which is an effective solution. Such performance enables simultaneous streaming of surgical video and patient vitals without delays.

4.5 Post-channel estimation constellation diagram

The post-channel estimation constellation diagram, as shown in Fig. 6, obtained at the output of the OFDM demodulator at the receiver side of the 4QAM OFDM LiFi system, exhibits a well-clustered signal, confirming that channel estimation and equalization have effectively mitigated most distortions introduced by the LiFi channel. The constellation points maintain their correct quadrant positions but exhibit minor scattering patterns, which might result from residual noise and multipath fading.

The noise in the optical amplifier and the cumulative attenuation of the channels in the 4QAM constellation cause scattering, and not the nonlinearity of the LED, which is not explicitly modelled in the study. In realistic VLC transmitters, LED compression usually starts at 60–80% of maximum drive current, and it will be considered in the future hardware validation to model realistic performance.

Signal quality in optical wireless communication will deteriorate from three primary sources, including LED non-linearity, optical turbulence, and ambient light interference. The quadrants remain properly separated even after challenges were addressed, thus leading to a better signal-to-noise ratio (SNR) while decreasing bit error rate (BER). Reliable high-speed data transmission emerges from the system despite minor remaining distortions that produce some dispersion.

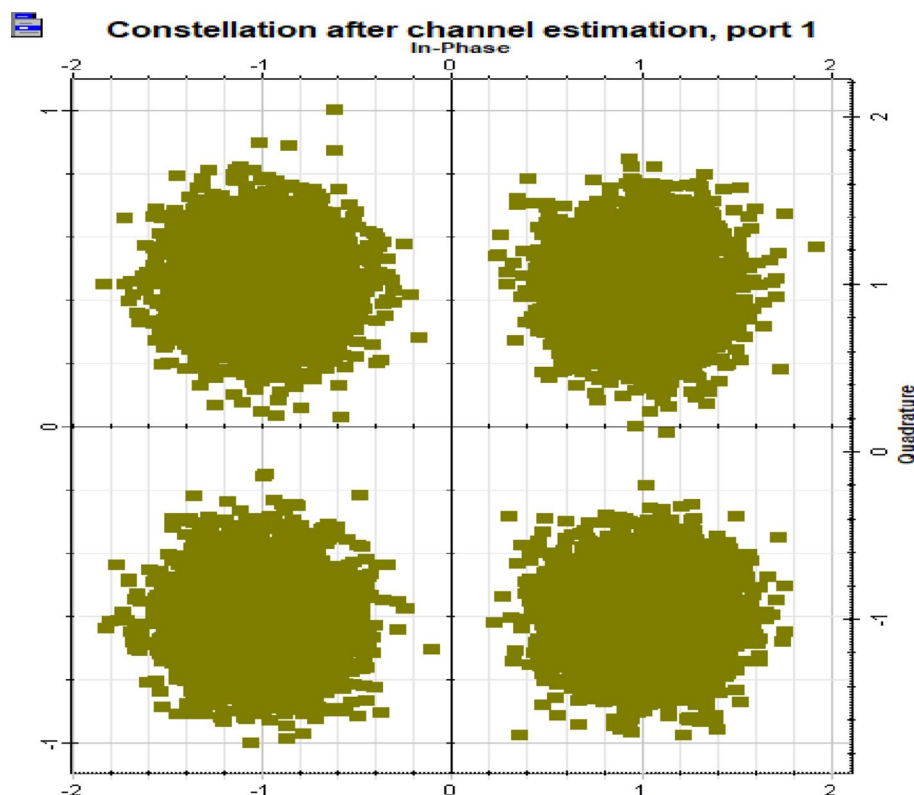


Fig. 6 Post-Channel Estimation Constellation Diagram at the OFDM Demodulator

In hospital applications, LiFi should deliver fully secure transmissions during real-time patient observation and medical data transfer operations. The confined nature of light-based transmission inherently reduces the risk of data leakage outside room boundaries, enhancing patient data privacy.

4.6 BER vs. distance analysis

In Fig. 7, the evaluation of BER versus distance at different OFDM power levels shows how transmission power affects performance in a LiFi-based hospital communication system.

The use of increased transmission power represents an effective method to lower BER, thus enhancing the reliability of hospital communication signals. The high BER at 15 dBm power level persists across all measured distances, which makes the communication system inappropriate for critical hospital applications. The transmission power set to 20 dBm performs better regarding BER, but the system quality still diminishes when signal strength exceeds short and medium distances. The selection of 30 dBm as the transmission power yields the best BER performance and superior error tolerance up to 15 m, which qualifies it as the best option for trustworthy hospital LiFi systems.

Research findings confirm the effectiveness of superior OFDM power levels to control path loss and ambient noise and interference, which results in low BER performance for critical real-time medical data transmission.

The optimal power level of OFDM (≥ 30 dBm) in a hospital setting proves vital because it maintains continuous connectivity for applications that include patient observation systems and AI diagnostic analysis, together with high-definition medical image processing.

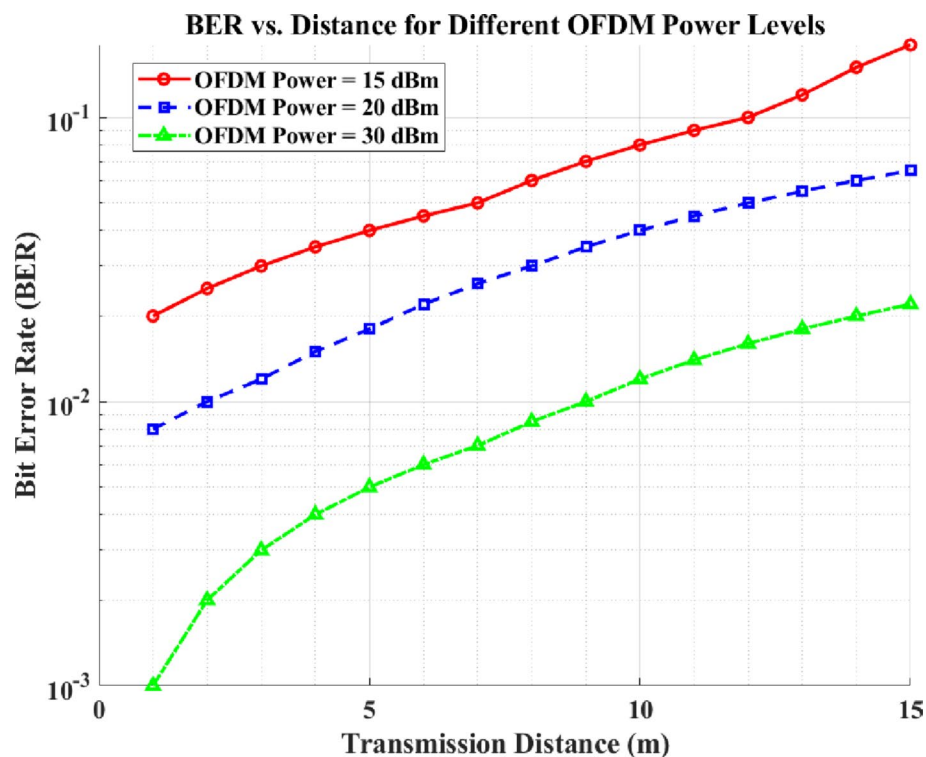


Fig. 7 BER vs. Distance for different OFDM levels

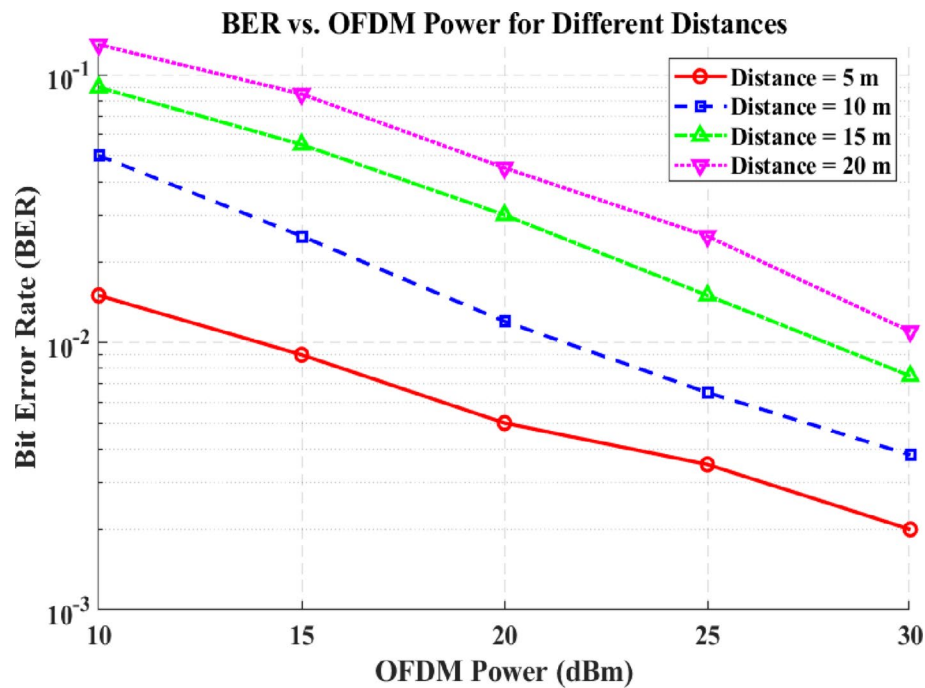


Fig. 8 BER vs. OFDM for different distances

4.7 BER vs. OFDM analysis

The BER versus OFDM power analysis for different transmission distances displays the impact of varying power levels on the error performance of a LiFi system in a hospital environment, as shown in Fig. 8.

An increase in OFDM power generates substantial BER reduction, which ensures signal reliability at every transmission range. The combination of transmission powers between 10 and 15 dBm produces high BER noise that severely affects signal reliability for distances beyond 15 m. When using 20–25 dBm power levels, the BER becomes better, but error rates surpass critical hospital application thresholds when measuring distances greater than 15 m. The measured results indicate that OFDM power amplification leads to lower BER values, which provides better signal reliability throughout every tested distance. The transmitting power set to 30 dBm delivers the optimal performance by achieving minimal error rates at a transmission distance of 20 m.

To benchmark the achievement of the proposed system in terms of standard-compliant BER when dealing with an optical wireless communication in an indoor setting, the VLC forward-error-correction (FEC) threshold established by IEEE 802.15.7 is 3.8×10^{-3} , and this should be used to evaluate the suitability of the proposed system in assuring the required standard-compliant BER.

Research findings indicate that a higher power level from OFDM transmission systems ensures low BER rates so that hospital LiFi networks can support secure and persistent communication needs of patient monitoring and other applications, including AI diagnostics and real-time medical imaging workflows.

All findings verify that the proposed 4QAM-OFDM LiFi system is an innovative design, aimed at hospitals, that possesses high spectral efficiency, high EMI immunity, and low BER characteristics, as compared to conventional RF and former LiFi systems. The combination of photobiologically secure 500 nm LED transmission, dense

OFDM-QM modulation, and optical amplifier optimization makes the system secure and allows transferring data at a within 15–20 m, which is appropriate to take part in essential applications like real-time diagnostics, surgical video streaming, and AI-aided medical analysis. These results confirm that it can be the superior successor to current wireless solutions, and that it deserves consideration as a next-generation hospital-wide wireless communication standard.

5 Conclusion

This paper proposes and validates a novel 4QAM-OFDM LiFi system specifically designed for EMI-sensitive hospital networks, rather than re-analysing existing solutions. The simulation results validate the proposed system as a dependable, high-performance solution that achieves secure data transfer in both biomedical and hospital facilities. This is, to the best of our knowledge, the first hospital-focused LiFi design to integrate photobiologically safe wavelength selection, optimized OFDM parameters, and realistic hospital channel modelling into one cohesive architecture. A full link budget consisting of saturation limits, thermal limits, photodiode linearity and ocular safety limits will be introduced into hardware prototyping. Also, the entire optical power levels that will be used in the hardware phase will be checked about thermal exposure and ocular safety according to IEC 62,471. The 30 dBm of the simulations is just a stress-test; in practice, the system will be used in hospitals with much lower optical powers that are eye safe and do not violate the photobiological safety standards in the area. These parameters are based on actual measurements of the device and are not part of the present simulation-only study. Future hardware testing will be conducted to ensure real-world BER, SNR, and latency measurements are made at different ambient light, human movement, and reflective hospital surface conditions because these factors cannot be properly simulated within a simulation. The laboratory tests will be done to demonstrate EMC compliance with IEC 60601-1-2 once a prototype hardware is created. The approach that is currently in use is based on theoretical EMI immunity and not certified EMC measurements. During the hardware stage, a complete security threat model, such as leakage analysis, eavesdropping risk, and injection attacks, will be created. Existing outputs are concerned with the advantages of architectural security, which are inherent to LiFi confinement.

Even though this work is simulation-based, it is a relatively safe way to test the LiFi system performance in the conditions of the hospital EMI without risks to prototype implementation. Future work will include hardware validation (LED driver characterisation and photodiode bandwidth testing) and the creation of a small-scale LiFi testbed for conducting controlled demonstrations in a hospital-room setting.

The LED-based transmitter transmits signals at 500 nm wavelength to minimize tissue absorption and delivers safe operation in proximity to medical equipment with a –100 dBm noise floor level. Using a 20 dB optical amplifier in the LOS receiver enables stable signal transmission over 10 m. The PIN photodetector generates electrical signals from optical inputs through its 0.6 A/W responsivity, while maintaining low noise levels.

The quadrature demodulator demonstrates superior spectral efficiency for signal reconstruction by maintaining minimal signal distortion when receiving the transmitted data. The constellation analysis at the OFDM demodulator proves that channel estimation and equalization work effectively while delivering reliable LiFi communication

with enhanced signal quality and minimized BER for real-time medical purposes. Transmission power increase through OFDM analysis leads to improved error performance, which results in reduced BER values at greater distances between the transmitter and receiver. The received signal quality when using 30 dBm transmitted power with 4.25×10^{-3} BER exhibited a passing standard that could achieve E-9 levels with error correction methods.

The significance of this work lies in delivering a secure, EMI-immune, and sustainable alternative to WiFi for hospitals, supporting mission-critical applications such as robotic surgery, AI-assisted diagnostics, and real-time medical imaging. By achieving superior BER performance, safety compliance, and energy efficiency, this system demonstrates clear advantages over previously reported LiFi implementations in generic indoor settings.

6 Future research directions

The research has established robust possibilities for LiFi-based hospital communication networks; however, various implementation aspects require further examination. Researchers should investigate the implementation of multiple-input multiple-output (MIMO) LiFi systems to enhance data throughput capabilities and expand network possibilities. The following stage of this study will also involve the formulation of a full multi-cell LiFi implementation plan of hospital floors, such as room coverage maps, AP placement optimization, inter-cell handover schemes, and co-channel interference modelling. Such studies will involve ray-tracing software and optical channel measurements that are hardware verified and will be run at the prototyping and field-testing phase. Besides MIMO-LiFi, the work in the future will involve the real hardware testing of the implementation with green LED front-end and a complete receiver chain to test the performance of the BER, the latency and the EMI under the conditions of the real hospital lights. Adaptive modulation means will be tested in order to sustain the quality of the link in the presence of changing illumination, patient motion, and variations in reflective surfaces. The strategies of artificial intelligence-based resource management and handover will be also created to guarantee the smooth connectivity between the hospital rooms, enhance the spectral efficiency, and decrease the power usage. Such improvements will facilitate the implementation of scalable, mobility-aware, and intelligent LiFi networks in healthcare facilities in practice.

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Author contributions

Ajay Sharma designed the system architecture, performed simulations, analysed results, and prepared the manuscript. Lalit Garg supervised the research framework and contributed to conceptual development and manuscript refinement. Ahmad Atieh supported OptiSystem modelling, validation, and technical review. Peter A. Xuereb contributed to results interpretation, manuscript proofreading, and technical corrections.

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Data availability

The datasets generated and analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

One of the authors, A.A., is affiliated with Optiwave Systems Inc., which provided the software for the simulation OptiSystem 21 we used for the study presented in the paper. However, there is no competing interest to Optiwave.

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